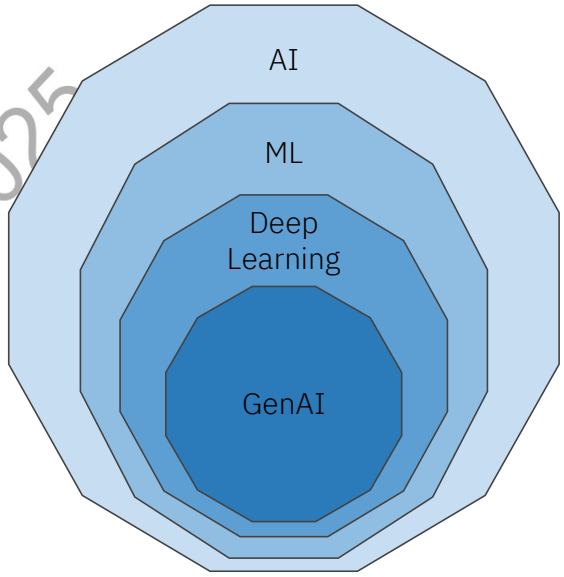
Introduction to AI

Week 5: Supervised machine learning

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Data & AI Leader, IBM ASEAN

Tran Dang An
Assistant

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1

Objectives

- Understanding of Machine learning and its categories: Supervised, Unsupervised and Reinforcement
- Supervised ML models: Linear Regression, KNN
- Labs: Spam Email

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2

2

AI Hierarchy

Artificial Intelligence

W1: Basic concepts, history of AI
W2: Knowledge representation and logic
W3: Searching algorithms (pathfinding and text)

Machine Learning

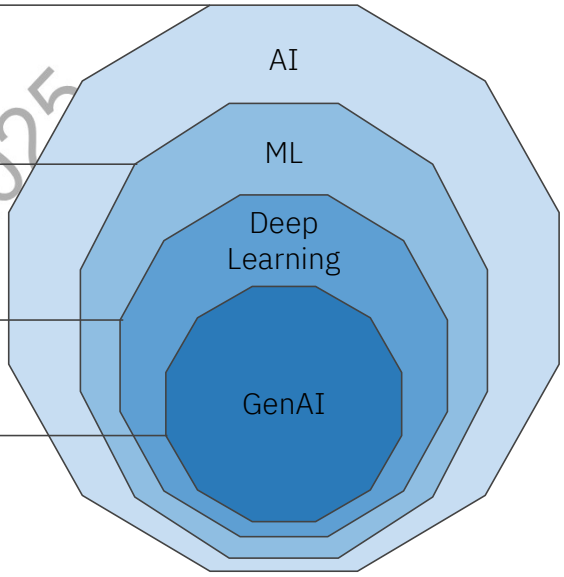
W4: Statistical learning
W5: Supervised machine learning
W6: Unsupervised machine learning

Deep Learning

W7: Perceptron learning and neural network learning
W8: Deep learning

Generative AI

W9 + W10: Modern AI and development in real world applications (LLM & GenAI)



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3

3

Contents

Machine Learning Overview
- Supervised Machine Learning
- Regression - Linear Regression
- Classification - KNN

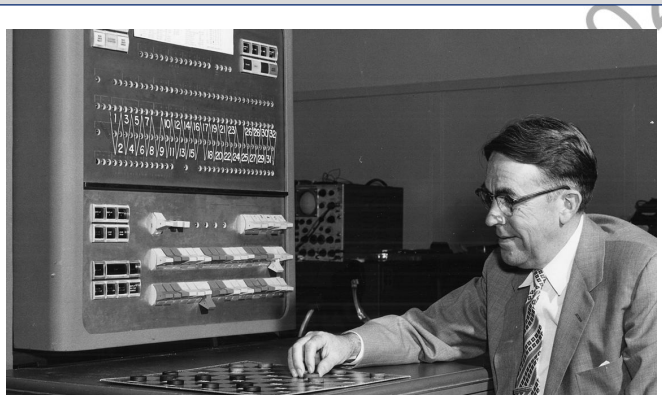
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4

4

Machine Learning definition

> "A field of study that gives computers the ability to learn without being explicitly programmed."
> *- Arthur Samuel (1959) -*



After joining IBM, the principles Samuel developed laid the groundwork for a series of breakthroughs in artificial intelligence at IBM during the 1990s

- **1952**: Arthur Samuel (IBM) builds a checkers program that learns from experience — an early ML system.
- **1992**: IBM's TD-Gammon uses neural networks + self-play to learn backgammon strategy.
- **1997**: IBM's Deep Blue beats the world chess champion, marking a major AI milestone.

5

5

From Statistical Learning to Machine Learning

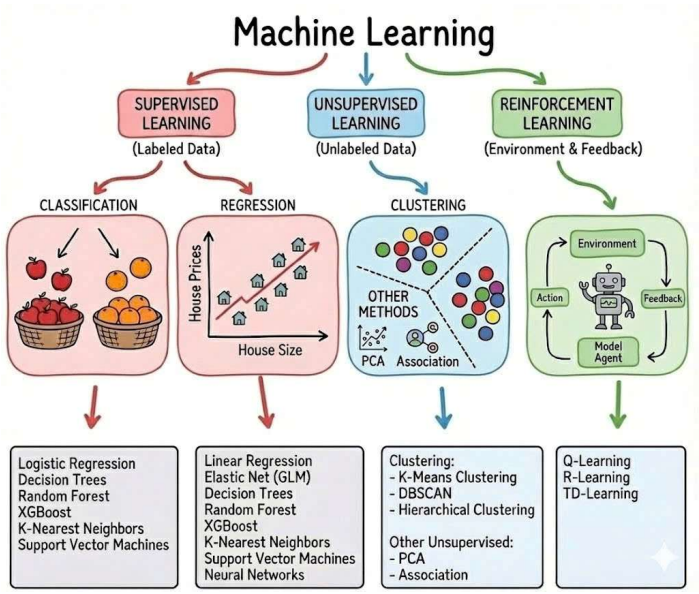
Statistical Learning	Machine Learning
• Moved beyond static rules to estimate mathematical relationships	• Computers learn and improve iteratively from examples.
• Used probability and variance to make sense of messy, real-world data.	• Uses statistical error minimization to find the "Best Answer" probability.
• Prioritized "Inference" to explain why relationships and patterns exist.	• Turns abstract math into concrete algorithms (e.g., Linear Regression, KNN).

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6

6

Categories of ML



7

7

Supervised vs Unsupervised (Use-cases)





Supervised

- Imagine a teacher showing a child a Red block and explicitly saying, "This is Red." or "This is Blue."

Unsupervised

- Imagine giving a child a bucket of mixed LEGO bricks but no instructions. The child might sort them just by looking at similarities.

8

8

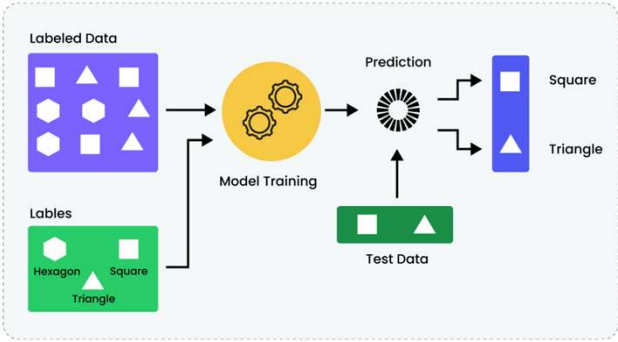
Supervised Learning

Formal Definition

"Supervised learning is a ML paradigm in which a model is trained on a labeled dataset consisting of input–output pairs (x_i, y_i) .The objective is to learn a mapping function $f: X \rightarrow Y$ that minimizes prediction error for unseen data."

Approach

- Training data contains **explicit labels** (ground truth), which the algorithm uses to adjust its parameters.
- Supervised learning aims to approximate the true underlying relationship between inputs and outputs.



Examples

- Classification: spam detection, dog/cat recognition
- Regression: predicting house prices, forecasting sales

Reference

The Elements of Statistical Learning (Hastie, Tibshirani, Friedman)

9

9

Unsupervised Learning

Formal Definition

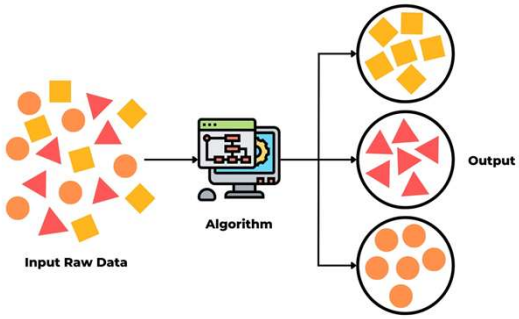
Unsupervised learning is a ML in which a model is trained on unlabeled data x_i ,with the objective of discovering hidden structure, patterns, or relationships within the dataset.

Approach

- No labels (y) are provided; the algorithm infer structure autonomously.
- The results are evaluated based on statistical properties, such clustering, feature learning or dimensionality reduction
- The goal may include clustering, dimensionality reduction, density estimation, or feature learning.

Examples

- Clustering: customer segmentation (K-means)
- Dimensionality reduction: PCA for visualization
- Anomaly detection: identifying outliers



Reference

Pattern Recognition and Machine Learning (Bishop)

10

10

Supervised vs Unsupervised (Definition)

Aspect	Supervised Learning	Unsupervised Learning
Data	Labeled (x, y)	Unlabeled (x)
Goal	Learn mapping f(x)→y	Discover structure in data
Evaluation	Prediction accuracy	Statistical quality (e.g., cluster cohesion)
Typical Methods	Logistic Regression, Trees, SVM	K-means, PCA, Autoencoders

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11

11

Introducing Loss function

Problem

- When a model makes a guess (Prediction $\hat{Y}$), it might be wrong.
- We need a mathematical way to calculate "How wrong is it?"

Approach

- The **Loss Function** is a formula that measures the distance between the Model's Prediction and the Actual Correct Answer *(for Supervised learning)*

The Rule:

- **High Loss:** The model is making bad guesses.
- **Low Loss:** The model is making accurate guesses.
- **Goal:** We want the loss to be as close to **zero** as possible.



The Loss Function: The amount of Static/Noise you hear. (More static = Bad model).

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12

12

Optimizing Loss function

Problem

- Knowing you are wrong isn't enough; you need to fix it.

Approach

- Loss optimization is the algorithmic process of adjusting the model to reduce the loss.

How:

- The optimization loop repeatedly tweaks the model's internal parameters (weights) to systematically minimize the Loss.



The Loss Function: The amount of Static/Noise you hear. (More static = Bad model).

Optimization: Turning the dial slightly left or right until the sound is clear.

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13

13

Key Terminology

The Foundation

- Features (X): The Inputs we feed the model (e.g., Number of Links).
- Labels (Y): The "Correct Answer" we want to predict (e.g., "Spam").
- Model (f): The mathematical engine that learns to map $X \rightarrow Y$.
- Dataset: The collection of all labeled examples used for training.

The Learning Mechanics

- Loss Function: The "Scorecard." A formula measuring error ($\hat{Y} - Y$). Goal: Minimize to 0.
- Parameters: The "Knobs." Internal variables (Weights) adjusted to improve accuracy.
- Optimization: The "Correction." The algorithm that updates Parameters to reduce Loss.
- Epoch & Convergence: One full pass through the data (Epoch) and the point where learning stops (Convergence).

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14

14

Contents

- Machine Learning Overview
- Supervised Machine Learning
- Regression - Linear Regression
- Classification - KNN

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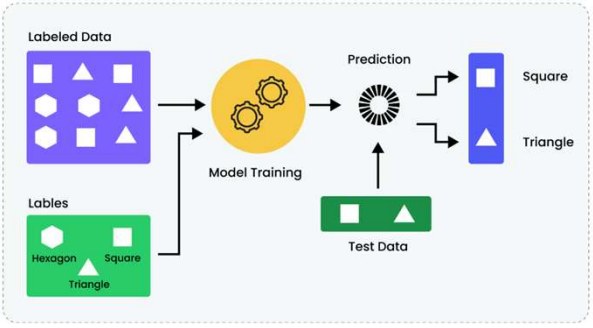
15

15

Supervised ML requires Labeled Data

To perform Supervised Learning, we strictly require a Labeled Dataset:

- **Features (Inputs *X*):** The data attributes used for prediction
- **Labels (Targets *Y*):** The correct answer or outcome for each example
- **Dataset**
  - **Training set**: Data with labels used to teach the model.
  - **Testing set**: Data held back to evaluate how well the model generalizes.



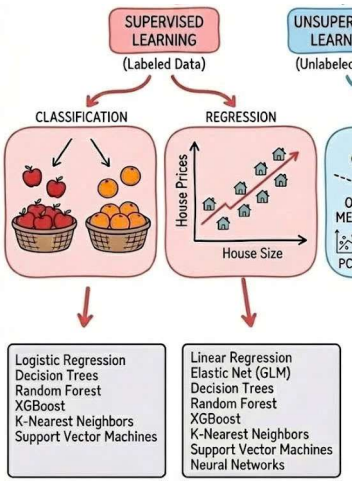
16

16

Classification vs Regression

These two sub-categories are chosen because, together, they introduce the two primary goals of Supervised Learning

- Classification:
  - Introduce Classification (predicting a label), using the highly intuitive concept of "neighbors voting" with almost no complex math.
- Regression:
  - Introduce Regression (predicting a number), demonstrating the concept of the "line of best fit" and measuring error.

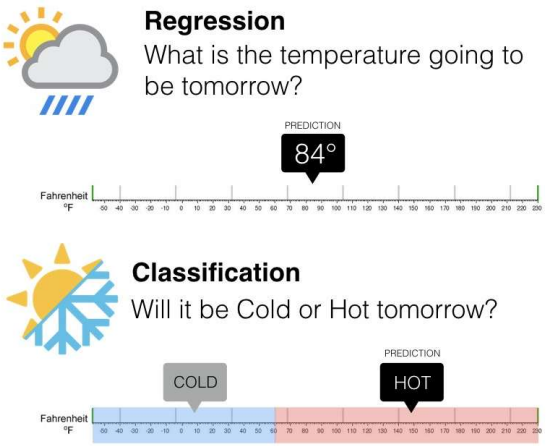


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17

17

Example



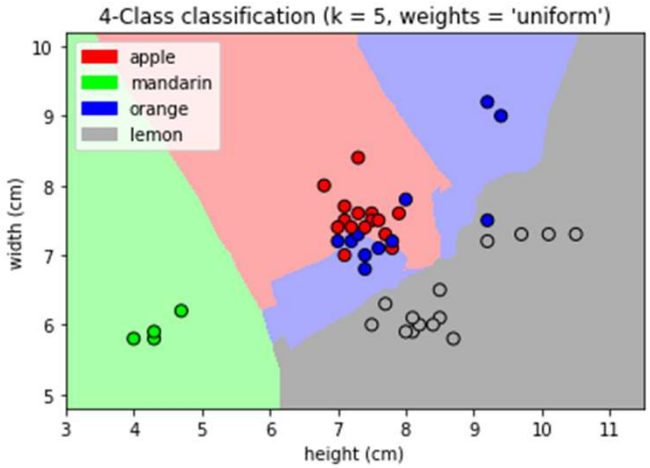
Source: Pintrest

18

18

Classification

Classification models predict a discrete label.



Real-World Examples:

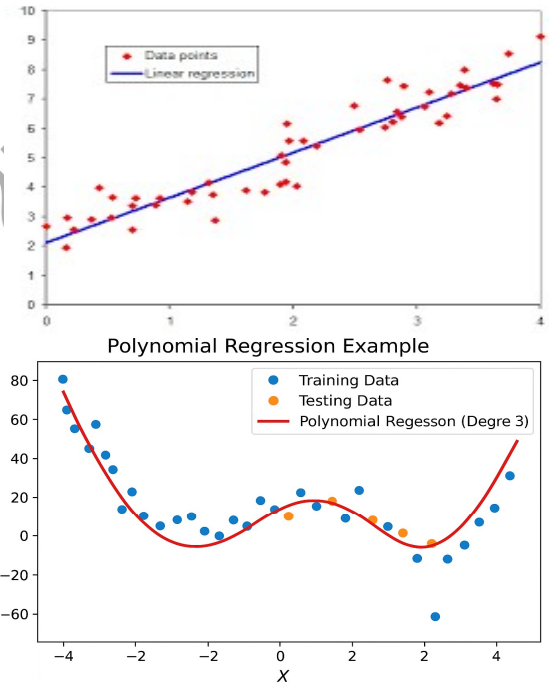
- Image Recognition: Is this a picture of a Cat or a Dog or Fox? (3 classes))
- Medical Diagnosis: Is the tumor non-cancerous (benign) or cancerous (malignant)? (2 classes)

19

19

Regression

Regression models predict a continuous number.



Real-World Examples:

- Real Estate: Predicting the exact price of a house
- Finance: Forecasting the stock price of Apple next week.

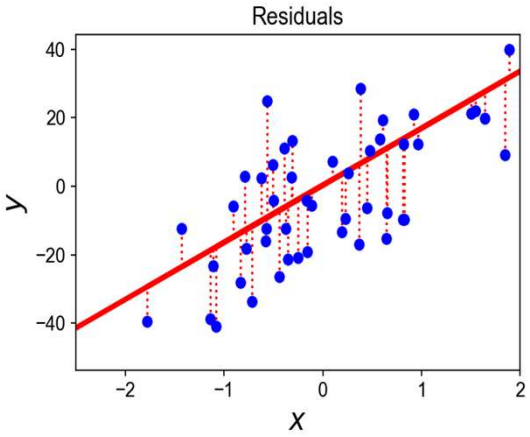
20

Loss Function

- **Prediction** $(\hat{Y})$: The value the model guesses.
- **Actual** (Y): The true value from our labeled data.
- **Loss Function**: A mathematical formula that calculates the difference between the prediction and the actual value.

$$Loss = \text{Difference between } \hat{Y} \text{ and } Y$$

- **The Objective**: The model's sole purpose is to minimize this as close to zero as possible.



One example of loss function using distance

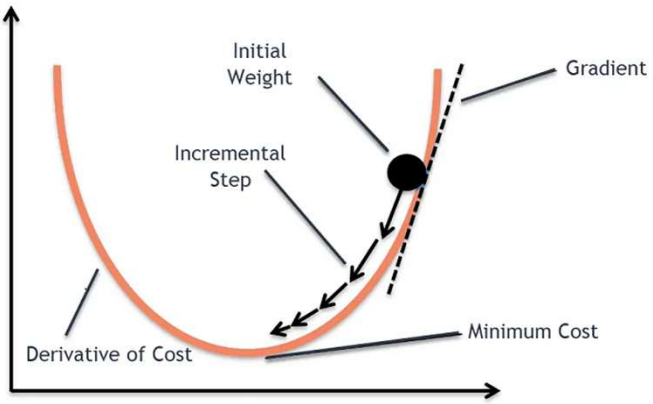
21

21

Loss Optimization: Gradient Descent Algorithm

Optimization is the mathematical process of adjusting the model's parameters to push the Loss value close to zero.

1. Find the lowest point of the valley (Minimum Loss), which represents the best model possible.
2. The algorithm checks the steepness (Gradient) and takes small steps downhill by adjusting the parameters.
3. It repeats this path until it hits the bottom (Convergence), meaning the error is minimized and training is complete.

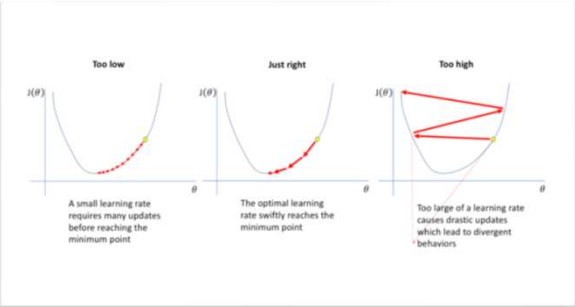


22

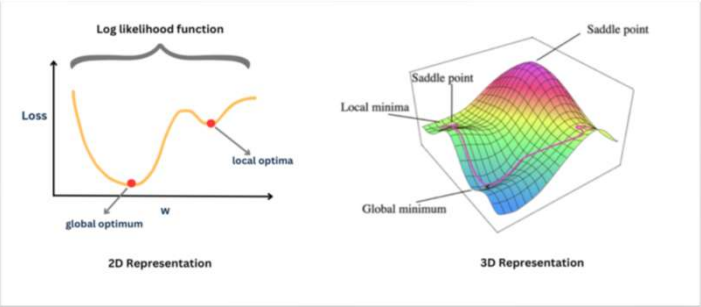
22

Other problems of Loss Optimization

Learning rate



Complex loss function and non-convex functions



https://machinemindscape.com/understanding-optimization-algorithms-in-deep-learning/

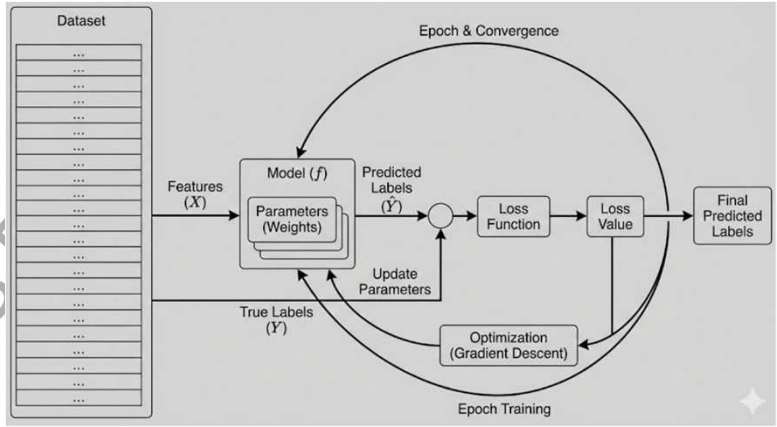
23

23

The Learning process

Learning is an iterative loop of optimization

1. **Forward Pass**: The model takes inputs (X) and makes a prediction ($\hat{Y}$).
2. **Calculate Loss (Error)**: It compares $\hat{Y}$ to the actual label **Y**.
3. **Tune Parameters**: The model adjusts its internal "weights" (mathematical settings) to reduce the error slightly.
4. **Repeat**: This cycle happens thousands of times until the error stops/slowly decreasing (Convergence).



24

24

Case Study: Spam Email

Predict spam email using **regression**
Problem: predict the Risk Score of email (Y)

Features X:
- Feature 1 (X_1): Number of Links (How many hyperlinks are in the email?)

The Model:
- Linear Regression: Predict the risk score (e.g., is it 80 or 90?).

Email ID	X1 (Links)	Regression Target (Y1)
#001	1	10 (Low Risk)
#002	5	45 (Medium Risk)
#003	20	95 (High Risk)
New	18	???

Predict spam email using **classification**
Problem: Classify the email (Y) as Spam or Not Spam

Features X:
- Feature 1 (X_1): Number of Links (How many hyperlinks are in the email?)
- Feature 2 (X_2): Number of Typos (How many spelling errors?)

The Model:
- KNN: Classify Category (Is it "Spam" or "Not Spam"?).

Email ID	X1 (Links)	X2 (Typos)	Classification Target (Y2)
#001	1	0	Not Spam
#002	5	2	Not Spam
#003	20	15	Spam
New	18	10	???

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25

25

Contents

- Machine Learning Overview
- Supervised Machine Learning
- Regression - Linear Regression
- Classification - KNN

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26

26

What Problem Does It Solve?

Linear Regression answers:

"How does Y change as X changes?"

Goal:
Find the line that best fits the data

Use cases:
- Predicting house price = f(size)
- Predicting exam scores from study hours
- Predicting temperature, revenue, sales, etc.



27

27

Definition

Linear Regression assumes a linear relationship:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Where:
- $\boldsymbol{Y}$: output (continuous)
- $\boldsymbol{X}$: input (feature)
- $\boldsymbol{\beta_0}$:intercept
- $\boldsymbol{\beta_1}$:slope
- $\boldsymbol{\epsilon}$: noise/error



28

28

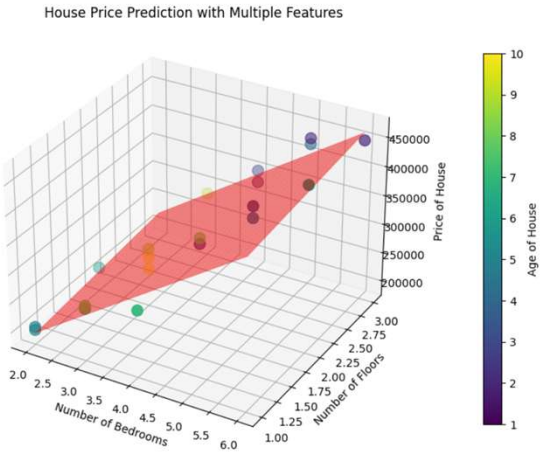
Multiple Linear Regression

When we have multiple features:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p + \epsilon$$

Examples:

- House price = f(size, bedrooms, floors)
- Sales = f(marketing spend, season, competitor price)



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29

29

How Linear Regression Learns (Distance-based)

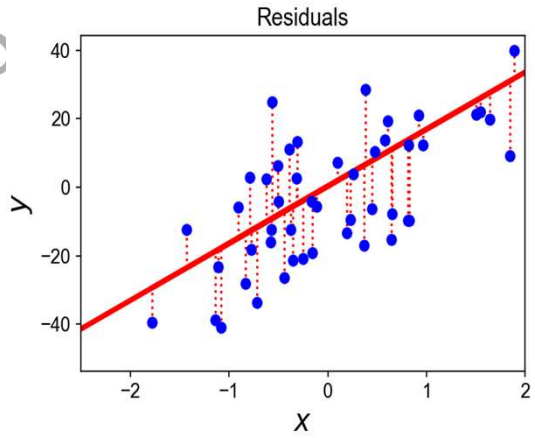
Residuals (Errors): vertical differences between points and the regression line

For each point:

$$\text{residual}_i = y_i - \hat{y}_i$$

- Positive residual
  → point is above the regression line
- Negative residual
  → point is below the regression line

Learning = Reducing These Residuals



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30

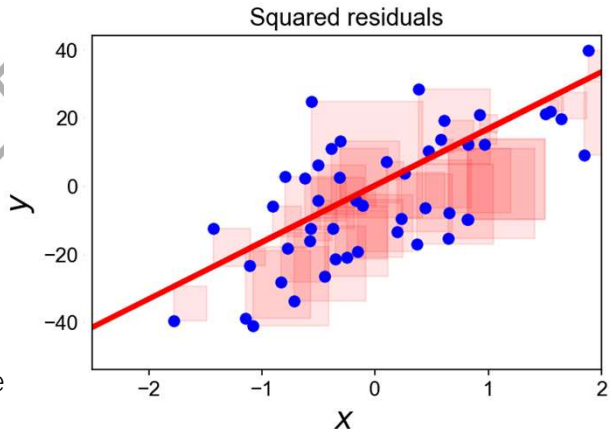
30

Loss Function: Squared Error

Linear Regression uses **Squared Error Loss**, which penalizes large errors more heavily:

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

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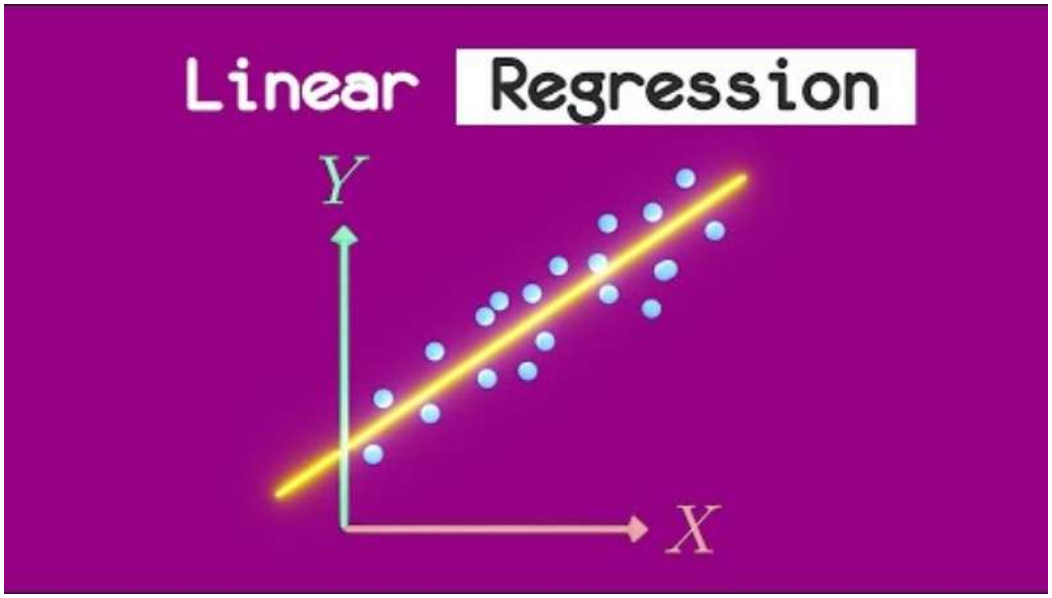
Why squared?

- Removes negative signs (all errors positive)
- Strongly penalizes large mistakes
- Gives a smooth, convex function → easy to optimize

31

31

Linear Regression Video

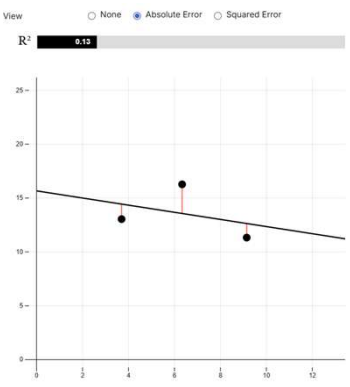


https://www.youtube.com/watch?v=CtsRRUddV2s

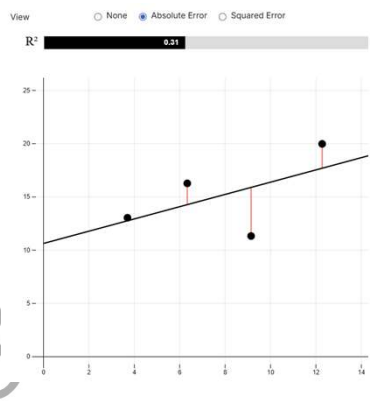
32

32

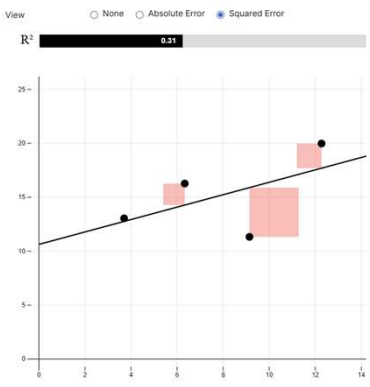
Linear Regression Simulation



Plot 3 data points



Plot the 4th data point



Change to squared error

https://observablehq.com/@yizhe-ang/interactive-visualization-of-linear-regression

33

33

Lab 5.1: Spam email with Linear Regression - Setup

- **Scenario**: Predicting the Spam Risk Score (0-100) based on email content.
- **Model Type**: Single Linear Regression ($1\ feature \rightarrow 1\ prediction$).
- **Selected Feature** (X): Number of Links (Hyperlinks in the email).
- **Prediction Target** (Y): Risk Score (The continuous danger level, 0-100).

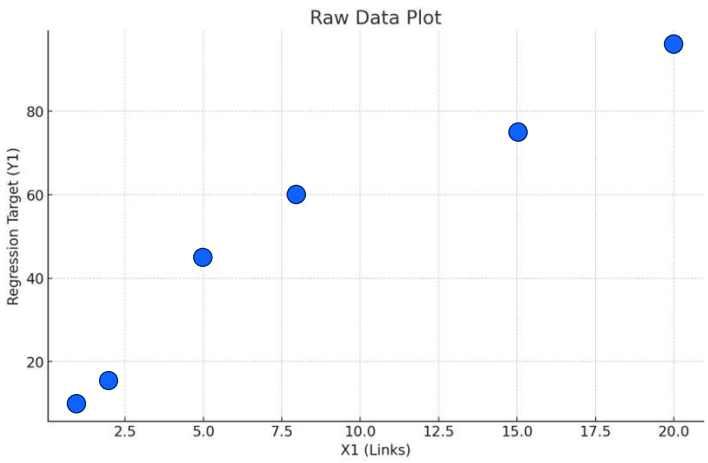
Email ID	X1 (Links)	Regression Target (Y1)
#01	1	10 (Low Risk)
#02	2	15
#03	5	45 (Medium Risk)
#04	8	60
#05	15	75
#06	20	95 (High Risk)
New	18	???

34

34

Lab 5.1: Data Visualization

- Plot the Number of Links on the X-axis and the Risk Score on the Y-axis.
- We can see that as the number of links increases; the risk score generally increases.
- Linear Regression find the line that best predicts this score based on the count of links.



35

35

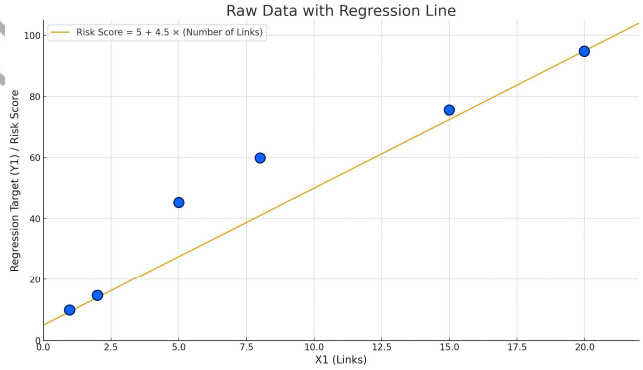
The Estimated Spam Function

- The Model: We are estimating the relationship: $Y = \beta_0 + \beta_1 X + \epsilon$
- The Model's Output $(\hat{Y})$: Our model learns the intercept $(\beta_0)$and the slope (β_1) that minimize error.
- Selected Function (by experience not by trained):

$$Risk\ Score = 5 + 4.5 \times (Number\ of\ Links)$$

  - Intercept $(\beta_0 = 5)$: If an email has 0 links, the starting risk is 5.
  - Slope $(\beta_1 = 4.5)$: Every additional link increases the risk score by 4.5 points.

Email ID	X1 (Links)	Regression Target (Y1)
#01	1	10 (Low Risk)
#02	2	15
#03	5	45 (Medium Risk)
#04	8	60
#05	15	75
#06	20	95 (High Risk)
New	18	???



36

36

Loss function: Calculate Residuals in Predicted Data

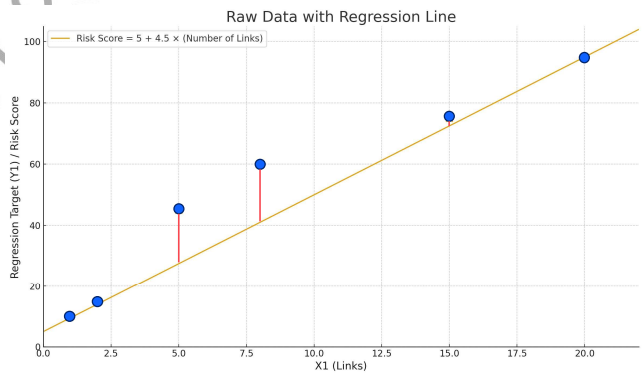
Loss function

- Calculate the residual for the email with 15 links
  *(it has an actual score of 75, but the line predicted 5 + 4.5*15 = 72.5)*

$$Residual = Y - \hat{Y} = 75 - 72.5 = 2.5$$

- Do the same for all other emails to get the total of residuals (using squared): ???

Email ID	X1 (Links)	Regression Target (Y1)
#01	1	10 (Low Risk)
#02	2	15
#03	5	45 (Medium Risk)
#04	8	60
#05	15	75
#06	20	95 (High Risk)
New	18	???

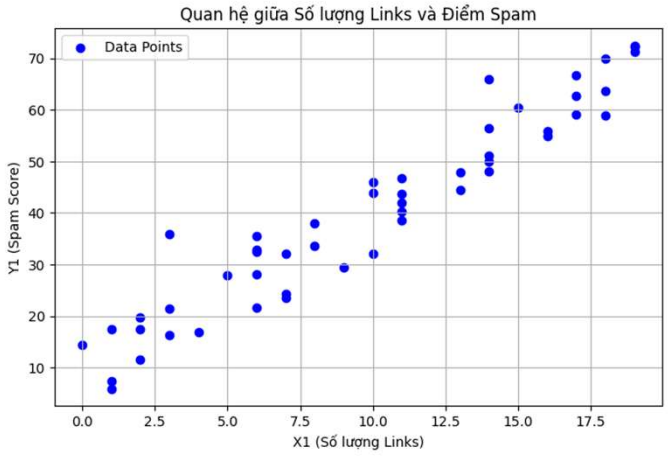


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37

37

Lab 5.1 Practice (45')



- Generate sample data
- Visualize data
- Setup the training: optimize the loss to get the final function

The entire dataset is used here to clearly illustrate the Gradient Descent process, allowing us to focus on the change in parameters (β_1, β_0) and the loss function over each epoch without added complexity.

Lab: https://colab.research.google.com/drive/1sGZANqhyFINDrFtr2O89yxMG7QaaWrXL?usp=sharing

38

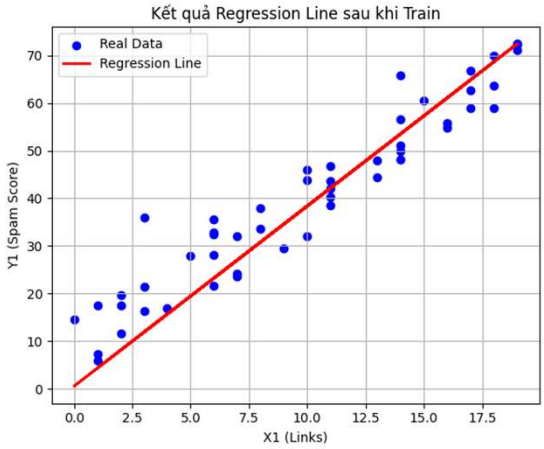
38

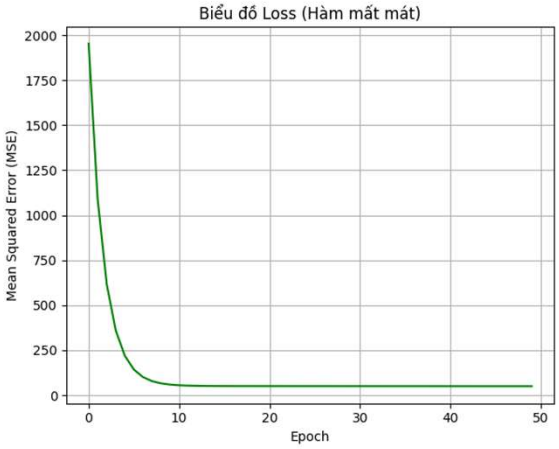
Lab 5.1 Linear Regression

Final Linear Function: y = 3.78x + 0.53

```
--- Bắt đầu Training (Gradient Descent) ---
Epoch 5: Loss = 219.38 | Line: y = 2.97x + 0.25
Epoch 10: Loss = 59.23 | Line: y = 3.62x + 0.32
Epoch 15: Loss = 51.40 | Line: y = 3.76x + 0.36
Epoch 20: Loss = 50.91 | Line: y = 3.79x + 0.38
Epoch 25: Loss = 50.78 | Line: y = 3.79x + 0.41
Epoch 30: Loss = 50.67 | Line: y = 3.79x + 0.43
Epoch 35: Loss = 50.55 | Line: y = 3.79x + 0.46
Epoch 40: Loss = 50.44 | Line: y = 3.79x + 0.48
Epoch 45: Loss = 50.33 | Line: y = 3.79x + 0.50
Epoch 50: Loss = 50.22 | Line: y = 3.78x + 0.53

Training hoàn tất!
Phương trình hồi quy cuối cùng: y = 3.78x + 0.53
```





39

39

Linear Regression— Key Takeaways

Pros

- Simple & Fast: Easy to train and highly interpretable.
- Inference: Tells us exactly how much a link affects the risk score (the slope β_1).

Cons

- It only predicts a number (the score).
- It cannot classify the email into a category (Spam vs. Not Spam) without adding an extra threshold rule.

Next, we will look at KNN, a model specifically built for classification

40

40

Contents

- Machine Learning Overview
- Supervised Machine Learning
- Regression - Linear Regression
- Classification - KNN

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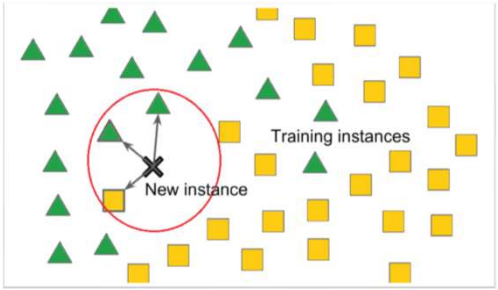
41

41

KNN Overview

Goal

A new data point is classified by assigning it the label most common among its nearest neighbors



An instance-based learning approach

KNN model does *not* learn a general formula during training.
Instead, it stores the training data (instances) and makes predictions by comparing new data to those stored examples.

Use cases:

- Image Recognition: Is this a picture of a Cat or a Dog?
- Medical Diagnosis: Is the tumor non-cancerous (benign) or cancerous (malignant)?
- Recommendation Systems: Classifying a new movie into a genre.

42

42

The Mechanism

1. **Training Data**: The model must store all the labeled training data (*X* and *Y*).
2. **Hyperparameter *K***: The number of neighbors to consider during the vote. K is set before training starts.
3. **Distance Metric**: The "closeness" between any two points to define “nearest neighbors”.

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43

43

How to measure the closeness between 2 data points?

Nobody has responded yet.

Hang tight! Responses are coming in.

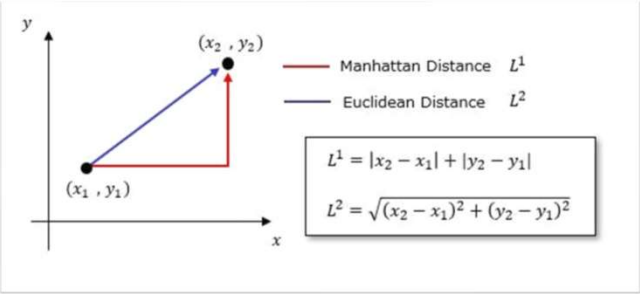
Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

44

Measuring Closeness w Euclidean Distance

Euclidean Distance (d): This is the length of the straight-line segment connecting the two points in a multi-dimensional space. It is to quantify how far a new, unclassified point p_1 is from any classified training point p_2

$$d(p_1, p_2) = \sqrt{\sum_{i=1}^{n} (p_{1i} - p_{2i})^2}$$



Where:

- p_1 and p_2 are the two points being compared.
- i is the feature index (e.g., $i = 1$ is Links, $i = 2$ is Typos).
- n is the total number of features (dimensions).

45

45

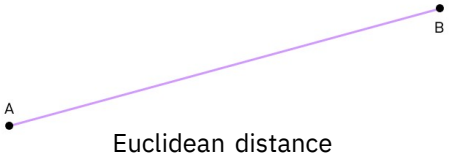
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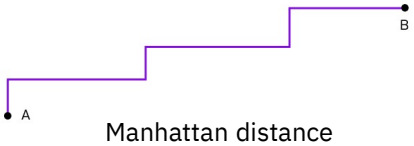
46

Other metrics to measure “distance” in ML

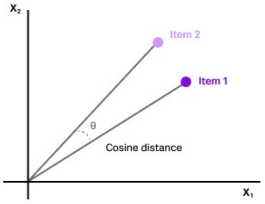
Distance metric: calculate the "closeness" between any two points



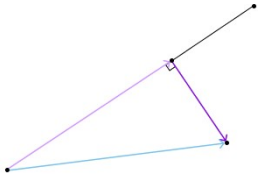
Euclidean distance



Manhattan distance



Cosine distance



Dot product

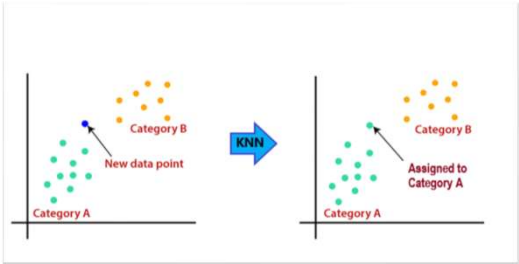
https://www.singlestore.com/blog/distance-metrics-in-machine-learning-simplfied/

47

47

KNN as Lazy Learner (no explicit training phase)

- **Model Type**: KNN is a "lazy learner," as it delays generalization until prediction time.
- **The Model is Memory**: The only "training" it performs is simply storing the entire labeled dataset in memory.
- **Prediction**: When a new point arrives:
  - The model measures the distance to every stored point
  - Selects the *K* neighbors
  - Uses their majority vote for the final label.
- **Trade-off**: This makes KNN fast to "train" but very slow for prediction on large datasets.



https://medium.com/@vishalkhomane1/lazy-algorithm-knn-regression-5e6b32b5f7c7

48

48

K-nearest Neighbors (KNN) video

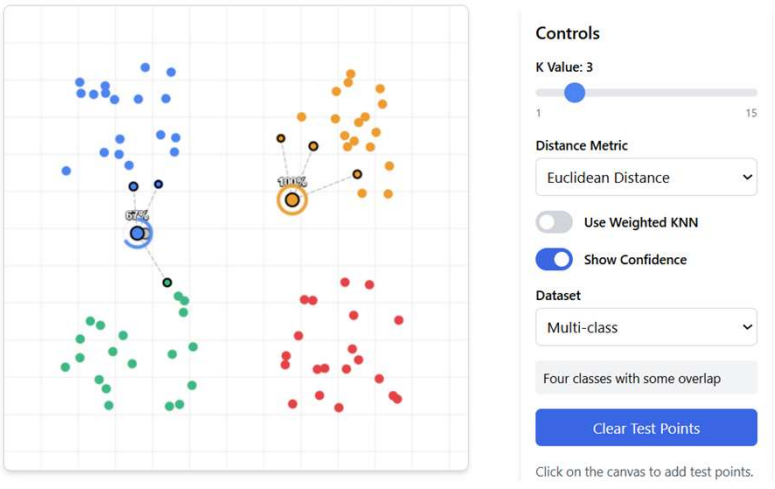


https://www.youtube.com/watch?v=gs9E7E0qOIc

49

49

K-nearest Neighbors (KNN) Simulation



https://www.luseratech.com/knn-visual

50

50

Lab 5.2: Spam Email with KNN - Setup

- **Scenario:** Classifying an email as Spam or Not Spam using the same data.
- **Selected Features (X):** Number of Links and Number of Typos (These become coordinates on a graph).
- **Prediction Target (Y):** Classification Label (Spam or Not Spam).
- **The Problem**: Plot our existing emails (Spam = Red Dots, Not Spam = Blue Dots). Where does the new email belong?

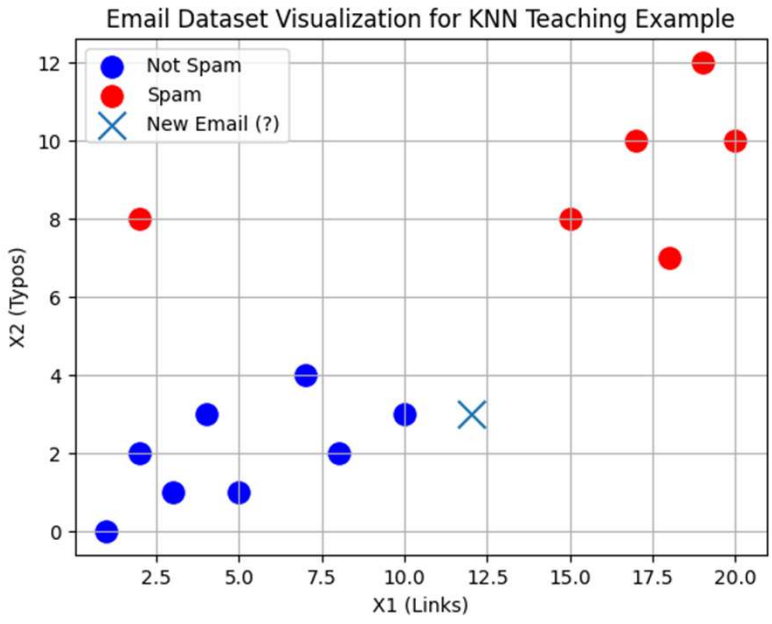
Email ID	X1 (Links)	X2 (Typos)	Y
#001	1	0	Not Spam
#002	3	1	Not Spam
#003	5	1	Not Spam
#004	2	2	Not Spam
#005	4	3	Not Spam
#006	7	4	Not Spam
#007	8	2	Not Spam
#008	10	3	Not Spam
#009	15	8	Spam
#010	18	7	Spam
#011	20	10	Spam
#012	19	12	Spam
#013	17	10	Spam
#014	2	8	Spam
New Point	12	3	???

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51

51

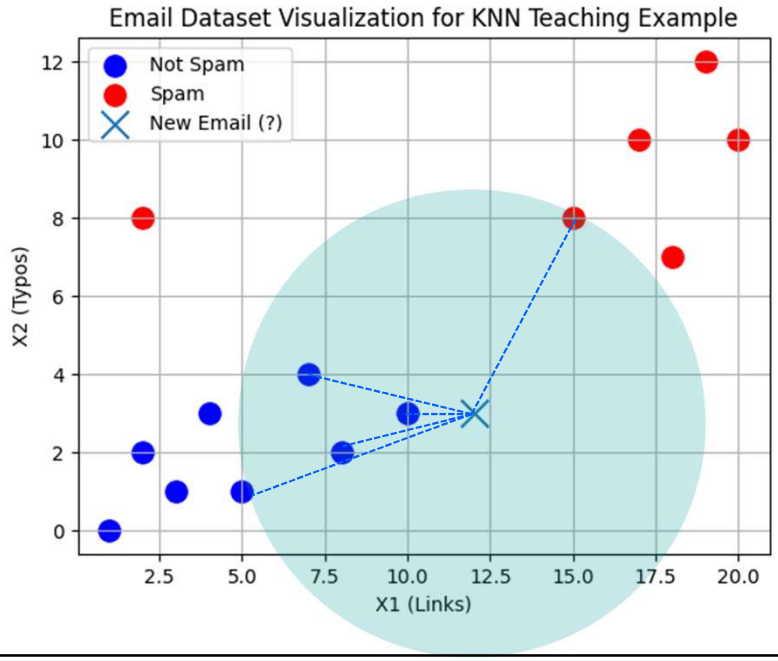
Lab 5.2: Data Visualization



52

52

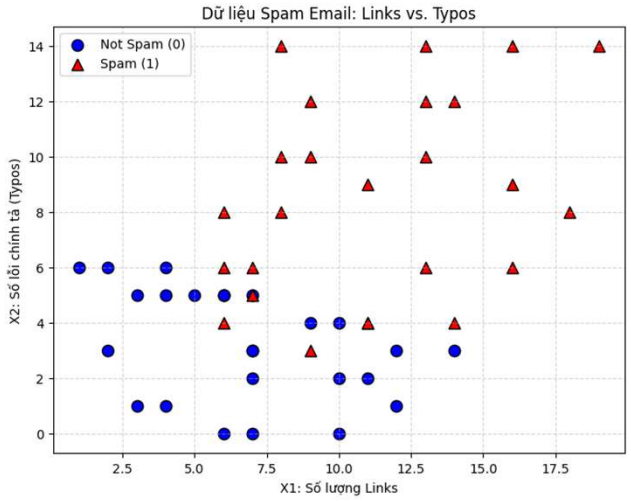
Lab 5.2: Making the Classification



53

53

Lab 5.2: Practices (45')



- Generate sample data
- Visualize data
- Setup the training
- Test

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Lab https://colab.research.google.com/drive/1dAqg9NAMsxyWCQb5YlDJemKVrQB-_aws?usp=sharing

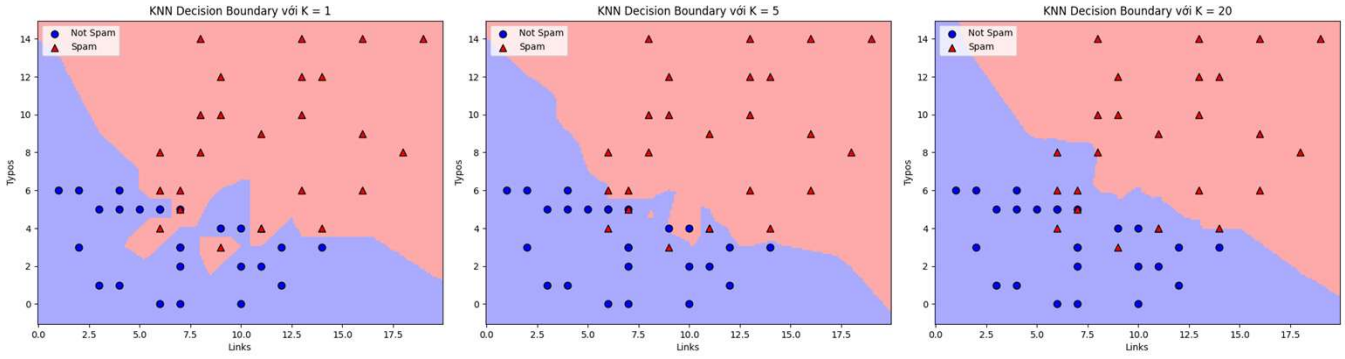
54

54

Lab 5.2: Practices with guidelines

- Train & test the model with different K values and spectate:
  - K = 1 -> Overfitting
  - K = 5 -> Optimized
  - K = 20 -> Too smooth, loss details (underfitting)

Complete the tasks in the notebook



55

55

Quiz to improve KNN

- Do we need to measure the distances to all points in the dataset?
  - Yes, as baseline KNN requires to do it for all points -> slow performance
- Any way to avoid that to improve the performance?
  - Divide data into group (tree) then just measure distances to points of the selected tree (K-dimensional Tree)
  - Select a suitable “zone” (Ball tree)
  - Select a suitable “zone” using a clustering like K-means (Cluster-based)
  - Don’t need to find the exact nearest neighbors — just an approximate result, but extremely fast (Approximate NN)

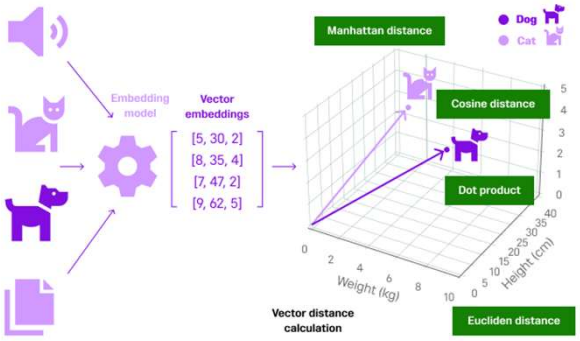
56

56

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Quiz

- Question: How to measure the closeness between Dog and Cat?
- Answer: We will use the vector distance, and we will learn in the GenAI lesson



57

57

KNN— Key Takeaways

Pros

- **Simplicity**: Highly intuitive and easy to explain—no complex math needed.
- **Non-Linearity**: Can model complex, non-straight boundaries better than Linear Regression.

Cons

- **Slow Prediction**: Must calculate the distance to every single data point to classify a new one. This makes it slow for large datasets.
- **Feature Scaling**: Sensitive to the scale of features (e.g., links vs. typos); features must be scaled equally.

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Next Steps: We have seen Regression (LR) and Classification (KNN).
We now have the foundation to explore more powerful algorithms!

58

58

Thank you!

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59

59

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